

In [14]:

```
1 import pandas as pd
2 df = pd.read_csv(r"C:\Users\Medida Sandeep\Data_Visualizations\data-viz-c
3
4 print("Original columns:", df.columns.tolist())
5
6 # Select correct columns explicitly
7 df_clean = df[[
8     'Country name',
9     'Ladder score',
10    'Logged GDP per capita',
11    'Social support',
12    'Healthy life expectancy',
13    'Freedom to make life choices',
14    'Generosity',
15    'Perceptions of corruption'
16 ]].copy()
17
18 # Rename properly
19 df_clean.columns = [
20     'Country',
21     'Happiness_Score',
22     'GDP',
23     'Social_Support',
24     'Life_Expectancy',
25     'Freedom',
26     'Generosity',
27     'Corruption'
28 ]
29
30 print(df_clean.head())
```

Original columns: ['Country name', 'Ladder score', 'Standard error of ladder score', 'upperwhisker', 'lowerwhisker', 'Logged GDP per capita', 'Social support', 'Healthy life expectancy', 'Freedom to make life choices', 'Generosity', 'Perceptions of corruption', 'Ladder score in Dystopia', 'Explained by: Log GDP per capita', 'Explained by: Social support', 'Explained by: Healthy life expectancy', 'Explained by: Freedom to make life choices', 'Explained by: Generosity', 'Explained by: Perceptions of corruption', 'Dystopia + residual']

	Country	Happiness_Score	GDP	Social_Support	Life_Expectancy	\
0	Finland	7.804	10.792	0.969	71.150	
1	Denmark	7.586	10.962	0.954	71.250	
2	Iceland	7.530	10.896	0.983	72.050	
3	Israel	7.473	10.639	0.943	72.697	
4	Netherlands	7.403	10.942	0.930	71.550	

	Freedom	Generosity	Corruption
0	0.961	-0.019	0.182
1	0.934	0.134	0.196
2	0.936	0.211	0.668
3	0.809	-0.023	0.708
4	0.887	0.213	0.379

```
In [15]: 1 import pandas as pd
2 df = pd.read_csv(r"C:\Users\Medida Sandeep\Data_Visualizations\data-viz-c
3 print("Original columns:", df.columns.tolist())
4
5 df_subset = df.iloc[:, :9] # Select only the first 9 columns
6 df_subset.columns = ['Country', 'Region', 'Happiness_Score', 'GDP', 'Social_S
7                      'Life_Expectancy', 'Freedom', 'Generosity', 'Corruption'
8
9 print(f"Dataset: {len(df_subset)} countries, {len(df_subset.columns)} col
10 print(df_subset.head())
```

Original columns: ['Country name', 'Ladder score', 'Standard error of ladder score', 'upperwhisker', 'lowerwhisker', 'Logged GDP per capita', 'Social support', 'Healthy life expectancy', 'Freedom to make life choices', 'Generosity', 'Perceptions of corruption', 'Ladder score in Dystopia', 'Explained by: Log GDP per capita', 'Explained by: Social support', 'Explained by: Healthy life expectancy', 'Explained by: Freedom to make life choices', 'Explained by: Generosity', 'Explained by: Perceptions of corruption', 'Dystopia + residual']

Dataset: 137 countries, 9 columns

	Country	Region	Happiness_Score	GDP	Social_Support	\
0	Finland	7.804	0.036	7.875	7.733	
1	Denmark	7.586	0.041	7.667	7.506	
2	Iceland	7.530	0.049	7.625	7.434	
3	Israel	7.473	0.032	7.535	7.411	
4	Netherlands	7.403	0.029	7.460	7.346	

	Life_Expectancy	Freedom	Generosity	Corruption
0	10.792	0.969	71.150	0.961
1	10.962	0.954	71.250	0.934
2	10.896	0.983	72.050	0.936
3	10.639	0.943	72.697	0.809
4	10.942	0.930	71.550	0.887

task1

```

In [16]: 1 !pip install plotly
2
3
4 import pandas as pd
5 import plotly.express as px
6 import plotly.graph_objects as go
7 region_avg = region_avg.sort_values('Happiness_Score')
8
9 fig1 = px.bar(
10     region_avg,
11     x='Happiness_Score',
12     y='Region',
13     orientation='h'
14 )
15
16 fig1.update_xaxes(range=[0, region_avg['Happiness_Score'].max()])
17
18 fig1.update_layout(
19     title="Western Europe Shows the Highest Happiness Levels Among Selected Countries",
20     plot_bgcolor='white'
21 )
22
23 fig1.show()

```

Requirement already satisfied: plotly in c:\users\j sree karthik\anaconda3\lib\site-packages (6.3.0)

Requirement already satisfied: narwhals>=1.15.1 in c:\users\j sree karthik\anaconda3\lib\site-packages (from plotly) (2.7.0)

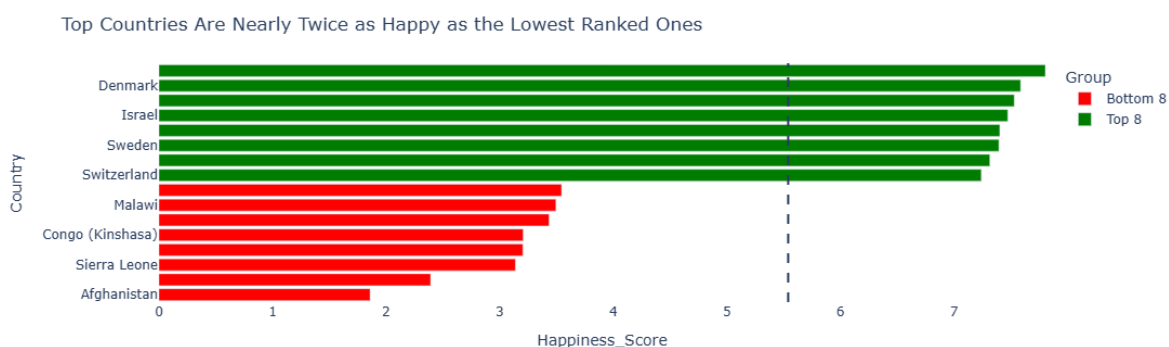
Requirement already satisfied: packaging in c:\users\j sree karthik\appdata\roaming\python\python313\site-packages (from plotly) (26.0)

Task-2

```

In [17]: 1 import pandas as pd
2 import plotly.express as px
3
4
5 top8 = df_clean.nlargest(8, 'Happiness_Score').copy()
6 top8['Group'] = 'Top 8'
7
8 bottom8 = df_clean.nsmallest(8, 'Happiness_Score').copy()
9 bottom8['Group'] = 'Bottom 8'
10
11 combined = pd.concat([bottom8, top8])
12 combined = combined.sort_values('Happiness_Score')
13
14
15 global_avg = df_clean['Happiness_Score'].mean()
16
17
18 fig = px.bar(
19     combined,
20     x='Happiness_Score',
21     y='Country',
22     orientation='h',
23     color='Group',
24     color_discrete_map={'Top 8': 'green', 'Bottom 8': 'red'}
25 )
26
27 fig.add_vline(x=global_avg, line_dash="dash")
28
29 fig.update_xaxes(range=[0, combined['Happiness_Score'].max()])
30
31 fig.update_layout(
32     title="Top Countries Are Nearly Twice as Happy as the Lowest Ranked Ones",
33     plot_bgcolor='white'
34 )
35
36 fig.show()

```



task-3

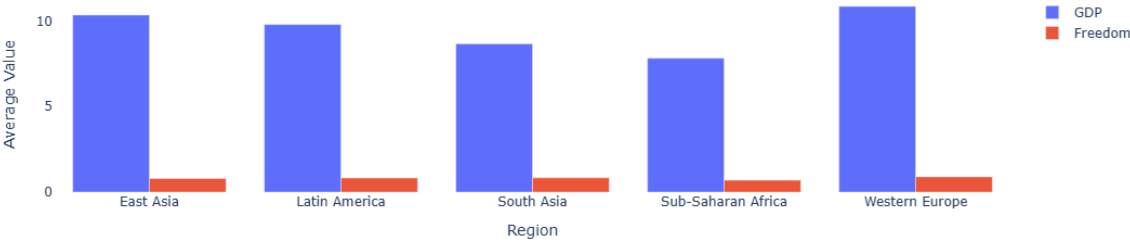
In [22]:

```
1 import pandas as pd
2 import plotly.graph_objects as go
3
4
5 if 'Region' not in df_clean.columns:
6     region_map = {
7         'Finland': 'Western Europe', 'Denmark': 'Western Europe', 'Iceland': 'Western Europe',
8         'Netherlands': 'Western Europe', 'Sweden': 'Western Europe', 'Norway': 'Western Europe',
9         'Germany': 'Western Europe', 'France': 'Western Europe', 'Spain': 'Western Europe',
10        'India': 'South Asia', 'Pakistan': 'South Asia', 'Bangladesh': 'South Asia',
11        'China': 'East Asia', 'Japan': 'East Asia', 'South Korea': 'East Asia', 'North Korea': 'East Asia',
12        'Nigeria': 'Sub-Saharan Africa', 'Kenya': 'Sub-Saharan Africa', 'Ethiopia': 'Sub-Saharan Africa',
13        'Congo (Kinshasa)': 'Sub-Saharan Africa', 'Sierra Leone': 'Sub-Saharan Africa', 'Liberia': 'Sub-Saharan Africa',
14        'Brazil': 'Latin America', 'Argentina': 'Latin America', 'Mexico': 'Latin America'
15    }
16    df_clean['Region'] = df_clean['Country'].map(region_map)
17    df_clean = df_clean.dropna(subset=['Region'])
18
19
20 regions = ['Western Europe', 'Latin America', 'East Asia', 'Sub-Saharan Africa']
21
22 df_clean.columns = df_clean.columns.str.strip()
23
24
25 required_cols = ['Region', 'GDP', 'Freedom']
26 for col in required_cols:
27     if col not in df_clean.columns:
28         raise ValueError(f"Column '{col}' not found in dataset")
29
30
31 subset = df_clean[df_clean['Region'].isin(regions)].copy()
32
33 subset[['GDP', 'Freedom']] = subset[['GDP', 'Freedom']].fillna(0)
34
35
36 grouped = subset.groupby('Region', as_index=False)[['GDP', 'Freedom']].mean()
37
38 print(grouped)
39
40 fig3 = go.Figure()
41
42 fig3.add_trace(go.Bar(
43     x=grouped['Region'],
44     y=grouped['GDP'],
45     name='GDP'
46 ))
47
48 fig3.add_trace(go.Bar(
49     x=grouped['Region'],
50     y=grouped['Freedom'],
51     name='Freedom'
52 ))
53
54 fig3.update_layout(
55     barmode='group',
56     title="Western Europe Leads in Both Economic Strength and Personal Freedom",
57     xaxis_title="Region",
58     yaxis_title="Average Value",
59     plot_bgcolor='white'
60 )
61
```

```
62 fig3.show()
```

	Region	GDP	Freedom
0	East Asia	10.349000	0.799333
1	Latin America	9.797000	0.826667
2	South Asia	8.661333	0.836000
3	Sub-Saharan Africa	7.818800	0.691600
4	Western Europe	10.853667	0.895556

Western Europe Leads in Both Economic Strength and Personal Freedom Compared to Other Regions



```
In [ ]:
```

1